

Namal Complaint Management System

Database Design Document

V 1.0

By

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1 PROJECT OVERVIEW

1.1 INTRODUCTION

The Namal Complaint Management System (NCMMS) is a web-based application designed to streamline the maintenance complaint management process at Namal University, Mianwali. The system aims to replace the current informal, verbal complaint reporting system by providing a centralized digital platform where students, faculty, and staff can submit maintenance-related complaints, track their status, and receive timely updates.

The system operates in a university environment where various types of maintenance issues arise, including electrical faults, plumbing problems, IT equipment failures, furniture damage, civil works, and HVAC system malfunctions. NCMMS provides role-based access for three primary user categories: End Users (students, faculty, staff who submit complaints), Maintenance Staff (personnel who resolve complaints), and Administrators (supervisors who manage the entire complaint lifecycle).

Key processes supported by the system include complaint submission with image attachments, automatic ticket generation, complaint assignment to appropriate staff members, real-time status tracking, work logging, resolution documentation, feedback collection, and comprehensive reporting. The system leverages modern web technologies including React.js for the frontend, Node.js/Express.js for the backend, MongoDB for data persistence, and Cloudinary for image storage.

1.2 PROBLEM STATEMENT

Namal University currently lacks a formal, digital system for managing maintenance complaints. The existing process relies on informal methods such as verbal reporting to hostel wardens, writing on physical notice boards, or sending messages through WhatsApp groups. This approach leads to several significant challenges:

Inefficient Complaint Tracking: Without a centralized system, complaints often get lost, forgotten, or delayed. There is no way to track whether a complaint has been acknowledged, assigned, or resolved.

Lack of Transparency: Complainants have no visibility into the status of their submitted complaints. They cannot determine who is handling their issue or when it will be resolved.

Poor Resource Management: Maintenance staff workloads are not systematically tracked. Some staff members may be overburdened while others have idle time, leading to inefficient resource utilization.

No Historical Record: There is no database of past complaints, resolutions, or recurring issues. This prevents the administration from identifying patterns, planning preventive maintenance, or evaluating staff performance.

Delayed Response Times: Without automated assignment and priority-based routing,

ing, critical complaints often experience the same response delays as minor issues.

Data Redundancy and Inconsistency: Information is scattered across multiple informal channels, leading to duplicate reporting and inconsistent records.

These problems result in decreased user satisfaction, prolonged resolution times, lack of accountability, and inability to make data-driven decisions for maintenance planning.

1.3 PROJECT OBJECTIVES

General Objective: To design, develop, and deploy a comprehensive database-backed complaint management system that automates the entire lifecycle of maintenance complaints at Namal University, from submission through resolution, while providing transparency, accountability, and analytical insights.

Specific Objectives:

- **Automate Complaint Lifecycle Management:** Develop a system that automatically generates unique complaint IDs, timestamps submissions, tracks status transitions, and maintains complete audit trails for every complaint.
- **Implement Role-Based Access Control:** Create secure authentication and authorization mechanisms supporting three distinct roles—End Users, Maintenance Staff, and Administrators—each with appropriate permissions and interfaces.
- **Enable Multi-Criteria Filtering and Search:** Provide users with the ability to filter and search complaints by status, category, priority, date range, and assigned staff member.
- **Provide Real-Time Status Updates:** Implement notification mechanisms (email and in-app) that automatically inform complainants when their complaint status changes or when comments are added.
- **Generate Analytical Reports:** Create reporting dashboards that display key performance metrics including average resolution time, complaint volume by category, staff performance, and recurring issue patterns.
- **Support Multimedia Attachments:** Enable users to upload images (up to 3 per complaint, max 5MB each) to better describe their issues and allow staff to document completed work with photos.
- **Ensure Data Integrity and Security:** Implement data validation, encryption for sensitive information, secure password storage using bcrypt hashing, and comprehensive audit logging for all critical operations.

1.4 GITHUB REPOSITORY LINK

https://github.com/SundeepKumar07/NAMAL_COMPLAINT_PORTAL

2 DETAILED DATABASE DESIGN

2.1 ENTITIES

Based on the requirements analysis and ERD modeling, the following core entities have been identified for the NCMMS database. These entities represent the primary objects, concepts, and events that must be tracked within the system.

Sr. No	Entity Name	Description
01	User	Represents every individual who interacts with the system. Stores authentication credentials, profile information, and role designation. Serves as the super-class for specialized user types.
02	EndUser	Specialization of User representing students, faculty, and staff who submit complaints. Contains department affiliation and user type classification.
03	MaintenanceStaff	Specialization of User representing personnel assigned to resolve complaints. Tracks availability status and current workload.
04	Admin	Specialization of User representing system administrators and supervisors with elevated access privileges.
05	Department	Represents academic and administrative departments within the university. Links EndUsers to their respective departments.
06	Category	Defines the classification hierarchy for complaint types including Electrical, Plumbing, IT, Furniture, Civil, and HVAC.
07	PriorityLevel	Defines urgency levels (Low, Medium, High, Critical) with associated Service Level Agreement (SLA) timeframes.
08	ComplaintStatus	Defines the lifecycle states for complaints including Open, Assigned, In Progress, Resolved, and Closed.
09	Building	Represents physical buildings on campus where complaints may originate.
10	Location	Represents specific rooms or areas within buildings where complaints are reported.
11	Complaint	The central entity representing a reported maintenance issue. Contains all submission details including title, description, timestamps, and foreign keys to related entities.
12	ComplaintAssignmentTrack	Tracks the assignment and reassignment history of complaints to maintenance staff.
13	WorkLog	Records work notes, actions taken, and time spent by maintenance staff on specific complaints.
14	Resolution	Captures final resolution details including summary and resolution timestamp for completed complaints.
15	ComplaintImage	Stores references to images uploaded by users or staff related to specific complaints.
16	Comment	Records textual comments added to complaints by any user type for communication and collaboration.
17	Notification	Manages system-generated notifications sent to users regarding complaint status changes and assignments.
18	StatusHistory	Maintains a complete temporal log of all status changes for each complaint.
19	Feedback	Stores user satisfaction ratings and comments after complaint resolution.
20	AuditLog	Records all critical system actions for security auditing and compliance purposes.
21	StaffSpecialization	Links maintenance staff to complaint categories, indicating areas of expertise.

2.2 DATA DICTIONARY

2.2.1 User Entity

The User entity serves as the base table for all system users, implementing a table-per-class inheritance strategy.

Sr. No	Name	Data Type	Constraint	Description
01	user_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique system-generated identifier for each user.
02	full_name	VARCHAR(100)	NOT NULL	Complete name of the user as per university records.
03	email	VARCHAR(100)	NOT NULL, UNIQUE	University email address used for login and notifications.
04	password_hash	VARCHAR(255)	NOT NULL	bcrypt hashed password for authentication.
05	phone_no	VARCHAR(20)	NOT NULL	Contact phone number for communication.
06	account_status	ENUM	NOT NULL, DEFAULT 'Active'	Current status: Active, Suspended, Deleted.
07	role_type	ENUM	NOT NULL	User role: EndUser, MaintenanceStaff, Admin.
08	profile_picture	VARCHAR(255)	NULL	URL path to user's profile picture stored in Cloudinary.
09	created_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Timestamp when user account was created.
10	last_login	DATETIME	NULL	Timestamp of user's most recent successful login.

Table 2. Data Dictionary: User Entity

2.2.2 Complaint Entity

The Complaint entity is the central table storing all complaint-related information.

Sr. No	Name	Data Type	Constraint	Description
01	complaint_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each complaint, visible as Ticket ID
02	title	VARCHAR(100)	NOT NULL	Brief summary of the complaint.
03	description	TEXT	NOT NULL	Detailed explanation of the issue.
04	submitted_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Timestamp when complaint was submitted.
05	updated_at	DATETIME	NULL, ON UPDATE CURRENT_TIMESTAMP	Timestamp of last modification to complaint record.
06	expected_resolution_date	DATETIME	NULL	Target date for complaint resolution based on priority SLA.
07	submitted_by	INT	NOT NULL, FOREIGN KEY	Reference to User who submitted the complaint.
08	category_id	INT	NOT NULL, FOREIGN KEY	Reference to Category entity.
09	status_id	INT	NOT NULL, FOREIGN KEY	Reference to ComplaintStatus entity.
10	priority_id	INT	NOT NULL, FOREIGN KEY	Reference to PriorityLevel entity.
11	location_id	INT	NOT NULL, FOREIGN KEY	Reference to Location entity.

Table 3. Data Dictionary: Complaint Entity

2.2.3 MaintenanceStaff Entity

The MaintenanceStaff entity extends the User entity with maintenance-specific attributes.

Sr. No	Name	Data Type	Constraint	Description
01	user_id	INT	PRIMARY KEY, FOREIGN KEY	References User.user_id.
02	availability_status	ENUM	NOT NULL, DEFAULT 'Available'	Current availability: Available, Busy, Off-Duty.
03	workload_count	SMALLINT	NOT NULL, DEFAULT 0	Number of active complaints currently assigned to staff.

Table 4. Data Dictionary: MaintenanceStaff Entity

2.2.4 Category Entity

The Category entity defines the classification system for complaint types.

Sr. No	Name	Data Type	Constraint	Description
01	category_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for complaint category.
02	category_name	VARCHAR(50)	NOT NULL, UNIQUE	Name of category: Electrical, Plumbing, IT, Furniture, Civil, HVAC, Other.
03	description	TEXT	NULL	Detailed description of what issues belong to this category.

Table 5. Data Dictionary: Category Entity

2.2.5 PriorityLevel Entity

The PriorityLevel entity defines urgency levels and associated SLA timeframes.

Sr. No	Name	Data Type	Constraint	Description
01	priority_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for priority level.
02	priority_name	VARCHAR(20)	NOT NULL, UNIQUE	Name: Low, Medium, High, Critical.
03	response_time_hours	INT	NOT NULL	Maximum hours within which complaint must be acknowledged.
04	sla_hours_resolution	INT	NOT NULL	Maximum hours within which complaint should be resolved.

Table 6. Data Dictionary: PriorityLevel Entity

2.3 BUSINESS RULES

The following business rules govern the behavior of the NCMMS and ensure data consistency, integrity, and correct system operation in alignment with university policies.

- **BR-01: User Registration and Verification** - Every user must register using a valid university email address. The system shall send a verification link to the registered email, and the user account shall remain inactive until verification is completed.
- **BR-02: Role-Based Access Control** - A user can have exactly one role (EndUser, MaintenanceStaff, or Admin). Permissions for all system operations shall be strictly enforced based on this role assignment.
- **BR-03: Complaint Submission Constraints** - A complaint can only be submitted by authenticated EndUsers. Each complaint must have a non-empty title (max 100 characters), description (max 1000 characters), valid category, priority level, and location. A maximum of three images (up to 5MB each) may be attached.
- **BR-04: Automatic Ticket ID Generation** - Upon submission, each complaint shall receive a unique ticket ID following the format NCMMS-YYYY-NNNN (e.g., NCMMS-2026-0001), where YYYY represents the year and NNNN is an auto-incrementing sequence number.
- **BR-05: Initial Complaint Status** - All newly submitted complaints shall have their status set to "Open". The system shall record this initial status in the StatusHistory table.
- **BR-06: Complaint Edit Window** - An EndUser may edit their complaint only if the complaint status is "Open" and the submission time is within 24 hours. After 24 hours or status change, editing is prohibited.
- **BR-07: Assignment Constraints** - A complaint may only be assigned to a MaintenanceStaff member whose StaffSpecialization includes the complaint's category. The system shall prevent assignment to staff members who are Off-Duty or have workload exceeding defined capacity (maximum 10 active complaints).
- **BR-08: Status Transition Rules** - Complaint status must follow the lifecycle: Open → Assigned → In Progress → Resolved → Closed. The system shall enforce valid transitions and record each change in StatusHistory. Status may revert from Resolved to In Progress if resolution is incomplete.
- **BR-09: Priority-Based SLA Enforcement** - The system shall calculate expected_resolution_date based on priority: Critical (24 hours), High (48 hours), Medium (5 days), Low (10 days). Notifications shall be triggered when a complaint approaches or exceeds its SLA deadline.

- **BR-10: Work Log Requirements** - Maintenance staff must add a WorkLog entry each time they update a complaint status. The entry must include a work note describing actions taken. When marking a complaint as Resolved, completion photos are encouraged but optional.
- **BR-11: Resolution Documentation** - When a complaint status changes to Resolved, the system shall automatically create a Resolution record capturing the resolution_summary (from staff comments), resolved_by (the staff member), and resolution_date (current timestamp).
- **BR-12: Feedback Collection** - After a complaint status changes to Closed, the system shall send a feedback request to the original complainant. A user may submit feedback only once per complaint, and feedback cannot be modified after submission.
- **BR-13: Deletion and Archival Policy** - Complaint records shall never be permanently deleted from the database. When an administrator deletes a complaint, the system shall set a soft-delete flag. Records older than 2 years shall be archived to secondary storage but remain accessible for reporting.
- **BR-14: Notification Rules** - The system shall send email notifications for the following events: complaint submission confirmation (to complainant), assignment (to assigned staff and complainant), status change (to complainant for In Progress, Resolved, Closed), and approaching SLA deadline (24 hours before expected resolution date).
- **BR-15: Audit Logging Requirements** - The system shall create an AuditLog entry for all CREATE, UPDATE, and DELETE operations performed on the Complaint, ComplaintAssignment, User, and PriorityLevel tables. Each log must capture the user, action type, affected table, timestamp, and a JSON snapshot of changes.
- **BR-16: Staff Workload Balancing** - When assigning a complaint, the system shall display the current workload count for each MaintenanceStaff member. Administrators are encouraged to assign complaints to staff members with the lowest active workload among those qualified for the category.

2.4 ENTITY RELATIONSHIP DIAGRAM

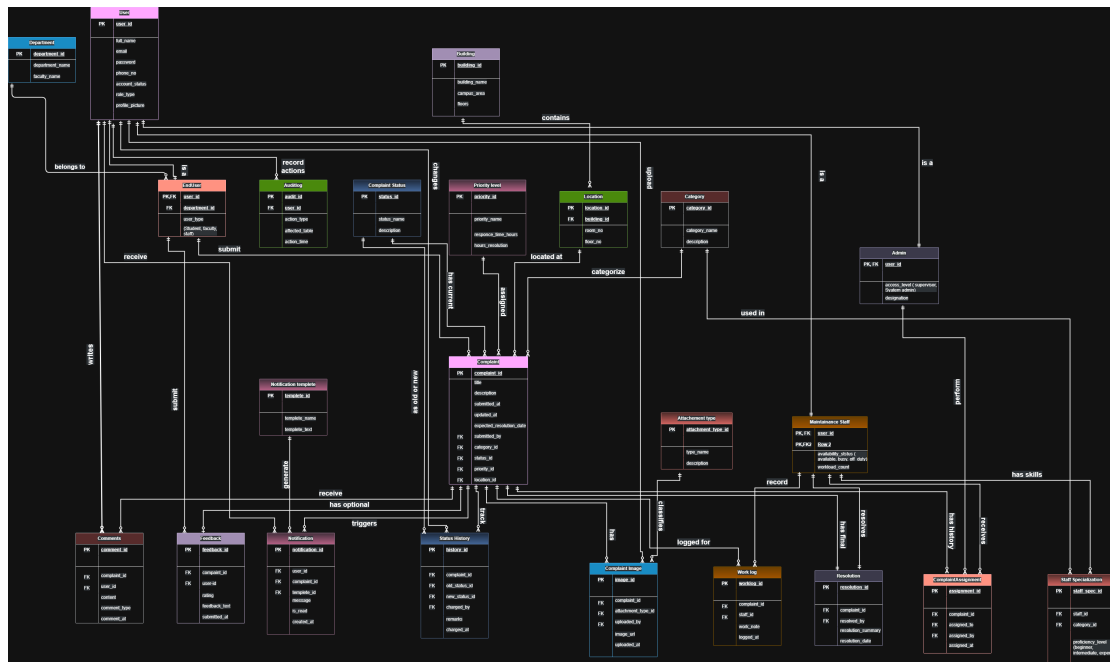


Figure 1. Entity-Relationship Diagram (ERD) - Namal Complaint Management System

2.4.1 Key Relationships and Cardinalities

Table 7. Summary of Relationships and Cardinalities

Relationship	Cardinality	Description
User $\rightarrow$ EndUser	1:1	Each User can be exactly one EndUser specialization (if role_type = 'EndUser')
User $\rightarrow$ MaintenanceStaff	1:1	Each User can be exactly one MaintenanceStaff specialization (if role_type = 'MaintenanceStaff')
User $\rightarrow$ Admin	1:1	Each User can be exactly one Admin specialization (if role_type = 'Admin')
EndUser $\rightarrow$ Complaint	1:M	One EndUser can submit many Complaints; each Complaint belongs to one EndUser
Complaint $\rightarrow$ Category	M:1	Many Complaints can belong to one Category; each Complaint has one Category
Complaint $\rightarrow$ PriorityLevel	M:1	Many Complaints can have one PriorityLevel; each Complaint has one PriorityLevel

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Relationship	Cardinality	Description
Complaint → ComplaintStatus	M:1	Many Complaints can have one current Status; each Complaint has one Status
Complaint → Location	M:1	Many Complaints can be located at one Location; each Complaint has one Location
Location → Building	M:1	Many Locations belong to one Building; each Location is in one Building
ComplaintAssignment → Complaint	M:1	Many Assignments can refer to one Complaint; each Assignment is for one Complaint
ComplaintAssignment → MaintenanceStaff	M:1	Many Assignments can be assigned to one MaintenanceStaff member
ComplaintAssignment → Admin	M:1	Many Assignments can be performed by one Admin
WorkLog → Complaint	M:1	Many WorkLog entries can belong to one Complaint
WorkLog → MaintenanceStaff	M:1	Many WorkLog entries can be recorded by one MaintenanceStaff member
Resolution → Complaint	1:1	Each Complaint has exactly one Resolution record; each Resolution belongs to one Complaint
Comment → Complaint	M:1	Many Comments can belong to one Complaint
Comment → User	M:1	Many Comments can be written by one User
ComplaintImage → Complaint	M:1	Many Images can belong to one Complaint
Notification → User	M:1	Many Notifications can be received by one User
Notification → Complaint	M:1	Many Notifications can be triggered by one Complaint
StatusHistory → Complaint	M:1	Many StatusHistory entries can track one Complaint
Feedback → Complaint	1:1	Each Complaint can have at most one Feedback record
StaffSpecialization → MaintenanceStaff	M:1	Many Specialization records can belong to one Staff member

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Relationship	Cardinality	Description
StaffSpecialization → Category	M:1	Many Staff members can be specialized in one Category

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APPENDIX: AI USAGE AND PROMPTS

This section documents the use of AI tools during the development of this Database Design Document.

Prompt Used	Purpose
"Generate a comprehensive data dictionary for a complaint management system including entities: User, Complaint, Category, PriorityLevel, Location"	Data dictionary generation for core entities
"Create business rules for a university complaint management system covering complaint lifecycle, assignment constraints, SLA enforcement, and audit requirements"	Business rules formulation
"Generate LaTeX code for a database design report based on the provided template structure with proper formatting for headings, tables, and figures"	LaTeX code generation for report structure
"Generate sample SQL/NoSQL schema definitions from the provided ERD for MongoDB implementation"	Schema implementation guidance

Table 8. AI Usage Log

AI Tool Used: DeepSeek (Latest version)

Usage Statement: The AI tool was used as an assistant to generate initial drafts of data dictionary entries, business rules, and LaTeX formatting code. All AI-generated content was reviewed, validated against the SRS document, and modified as necessary to ensure accuracy and compliance with project requirements. The ERD design, relationship cardinalities, and all technical decisions were made by the development team.